REVISION	DESCRIPTION	DATE	APPROVED

Project Name:

Projeto de Circuitos eletrônicos para IoT V1.0.0 Released 09/10/2024

Page	Contents
1	Table of contents
2	System block diagram
3	Top level hierarchical connections
4	Sensors
5	Board buttons and LEDs
6	RGB LED
7	Battery power management
8	Connectors
9	Mechanical components

Design Notes

Example of informational notes.

Example of cautionary design notes.

Example of critical design notes.

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PROJECT: Joab da Silva Bezerra- PCEIoT.PrjPcb	VERSION: V1.0.0
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DOCUMENT NAME: [1] Table of Contents.SchDoc	SHEET: 1 OF 9
DOCUMENT REVISION: Not in version control	
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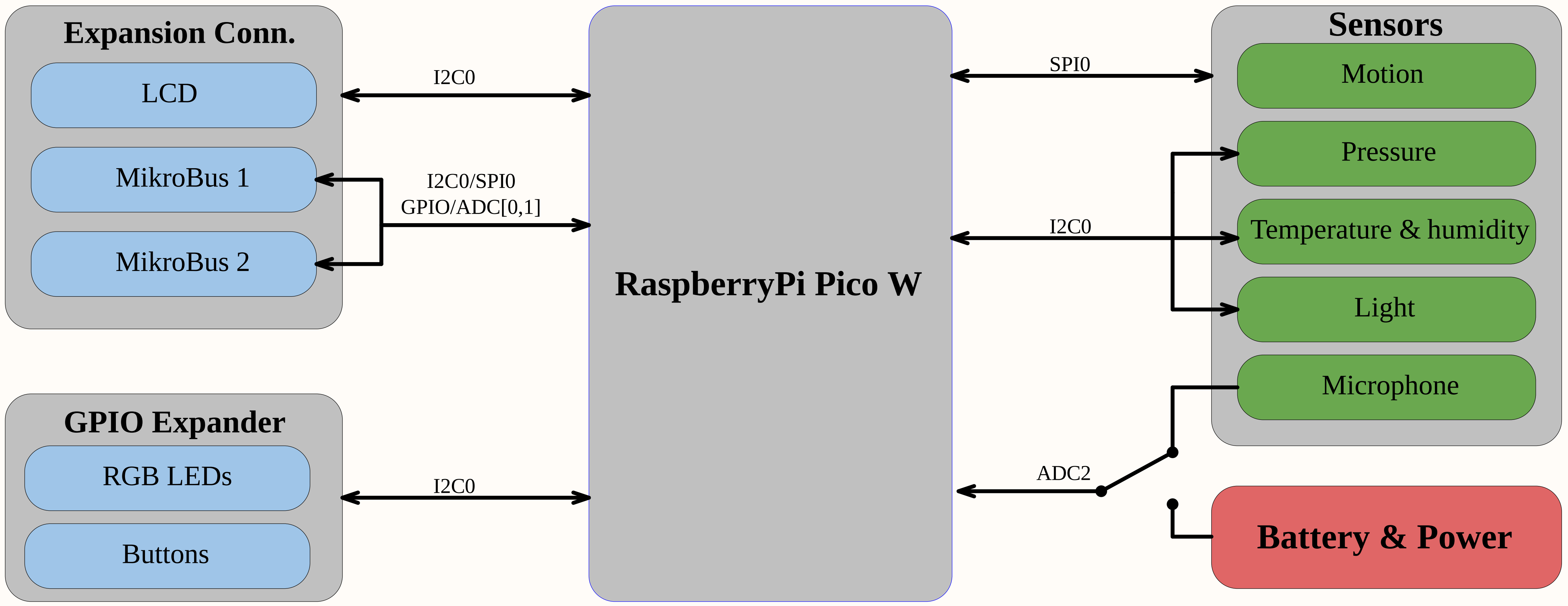

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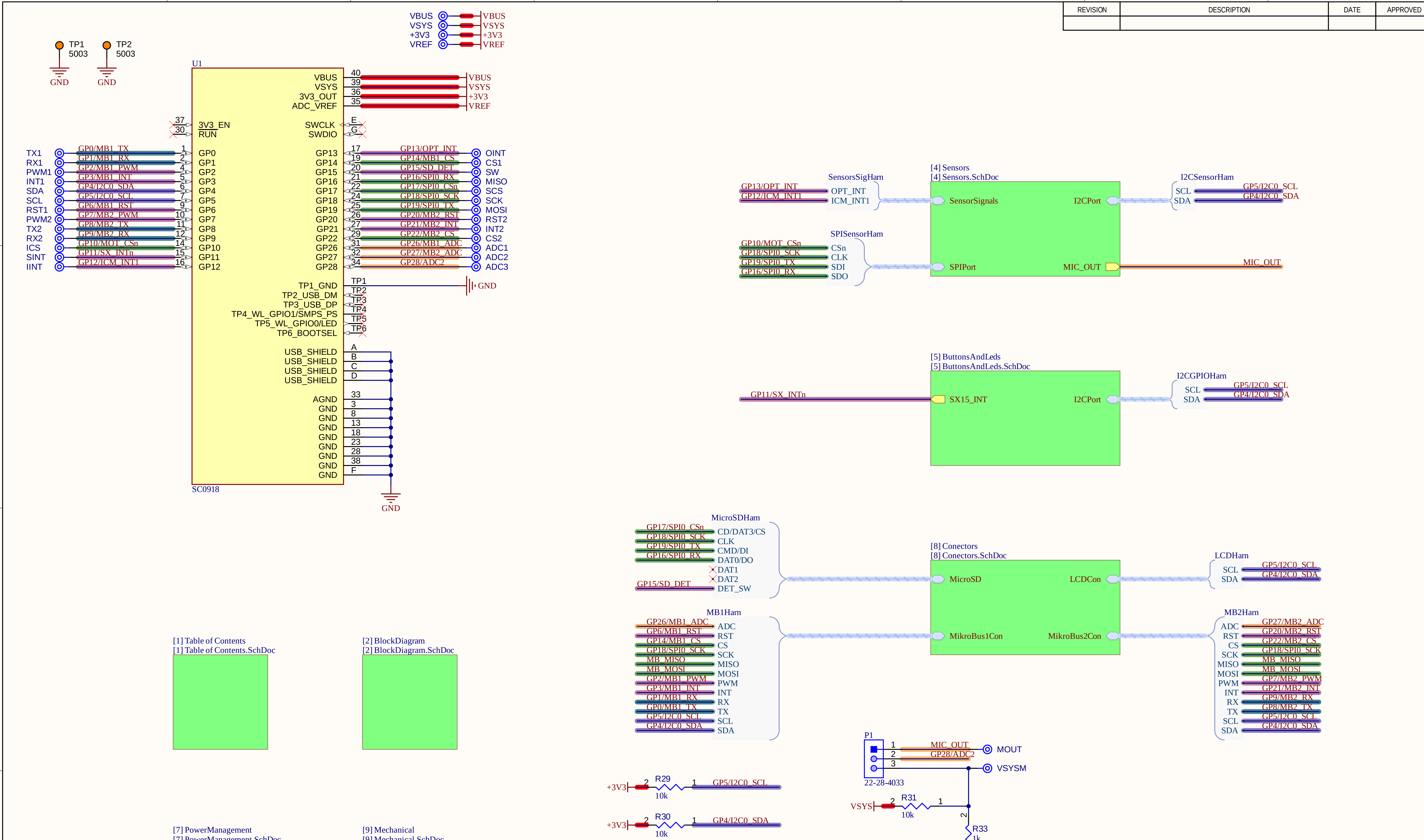

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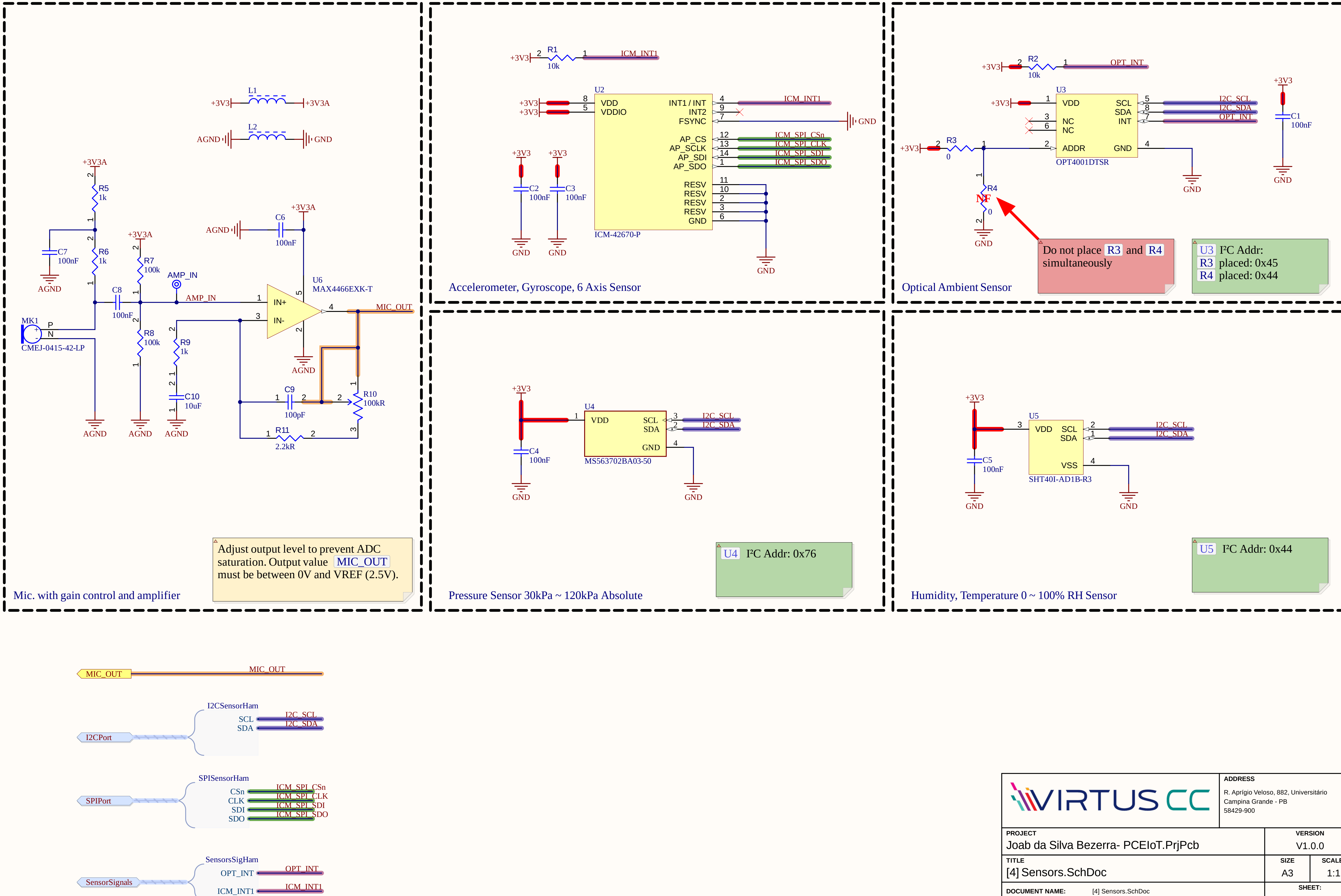

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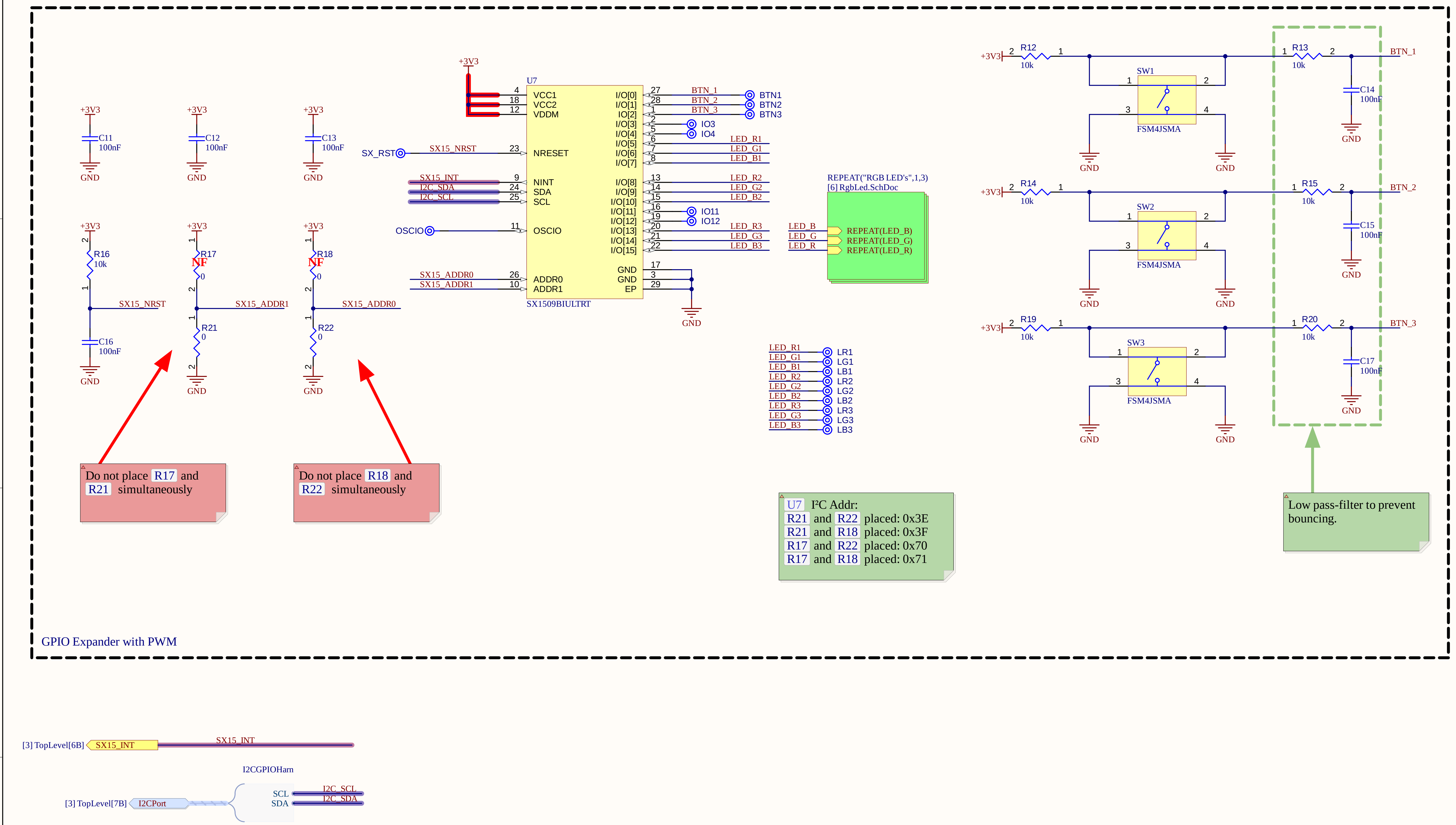

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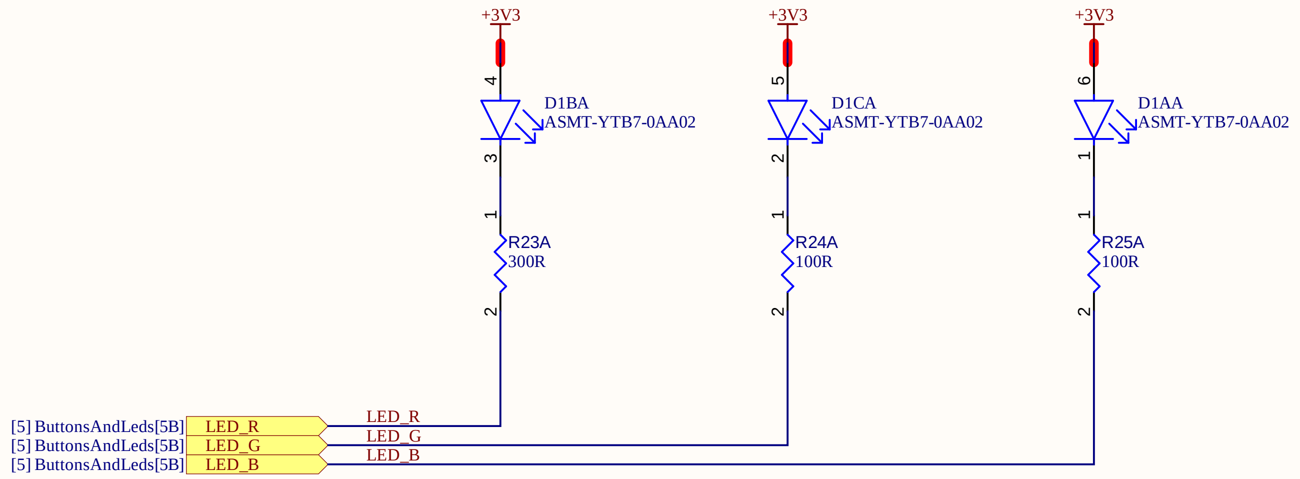

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PROJECT Joab da Silva Bezerra- PCEIoT.PrjPcb	VERSION V1.0.0
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DOCUMENT REVISION: Not in version control	6 OF 9
PROJECT REVISION: Not in version control	

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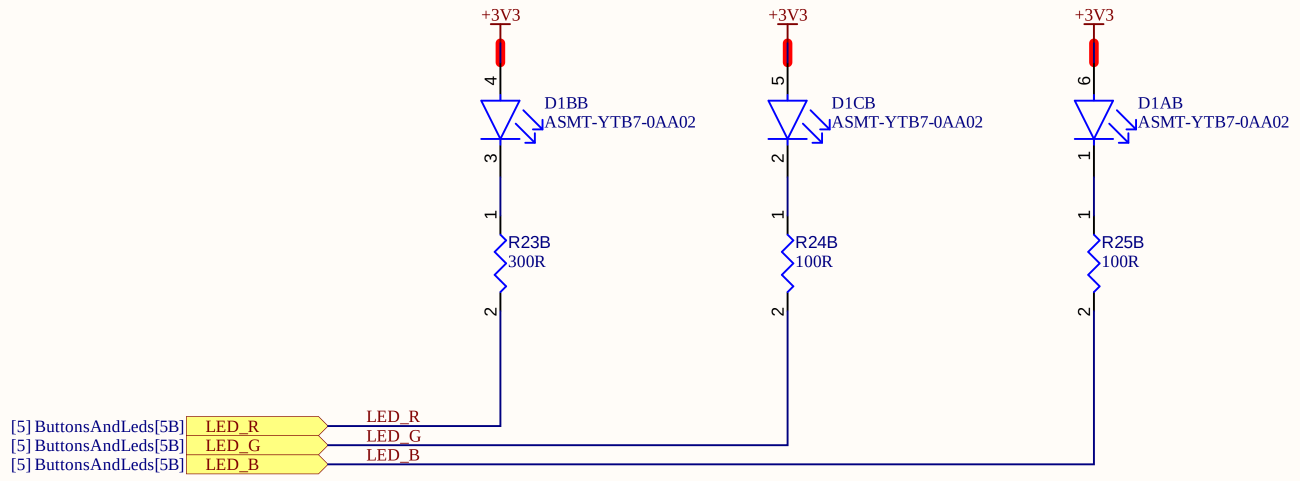

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PROJECT	Joab da Silva Bezerra- PCEIoT.PrjPcb	VERSION	V1.0.0
TITLE	[6] RgbLed.SchDoc	SIZE: A3	SCALE: 1:1
DOCUMENT NAME:	[6] RgbLed.SchDoc	SHEET:	6 OF 9
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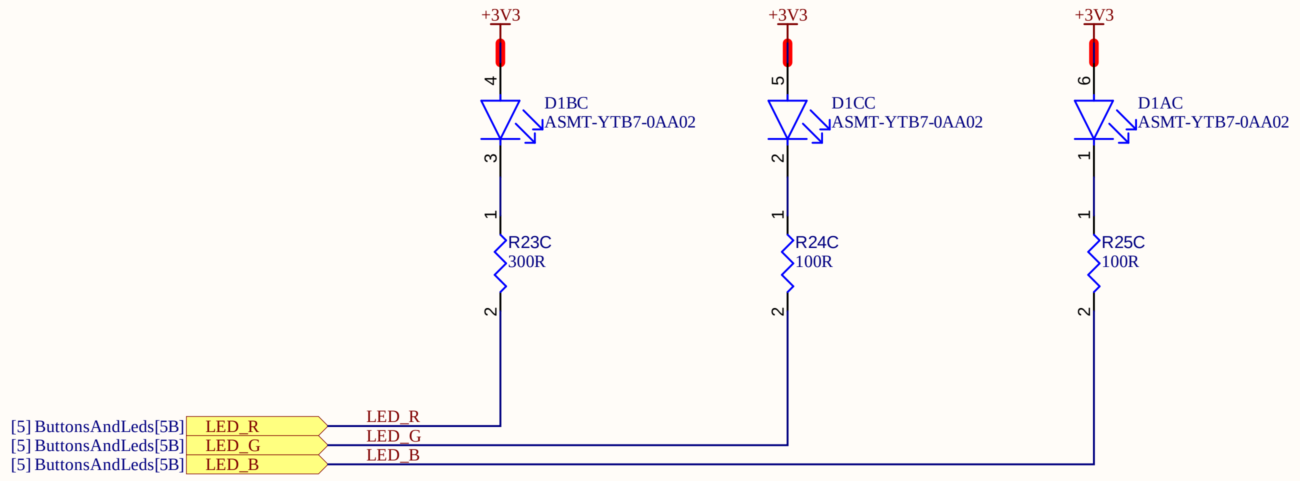

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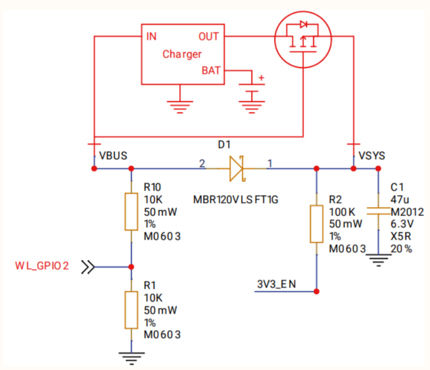
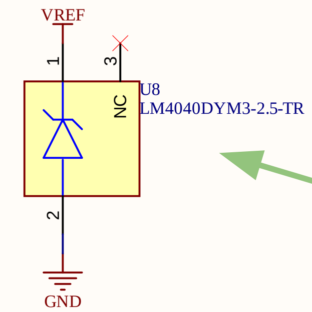
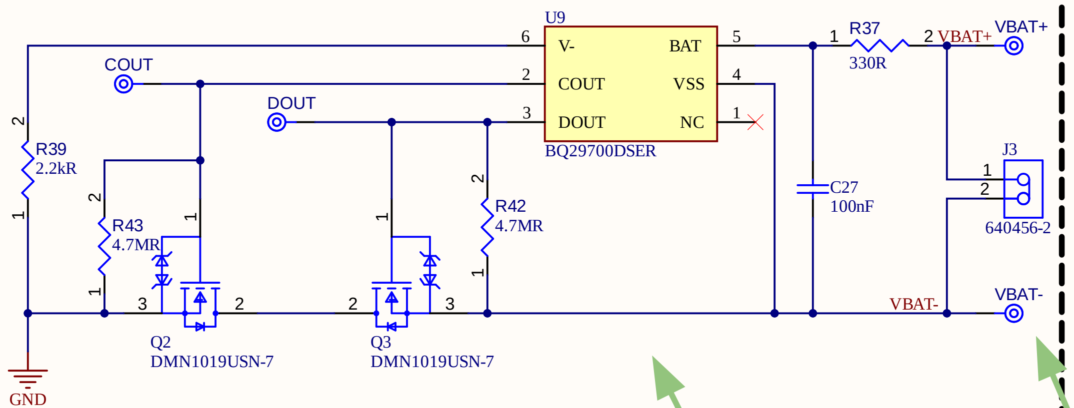


RPI Pico W Battery Suggestion Usage

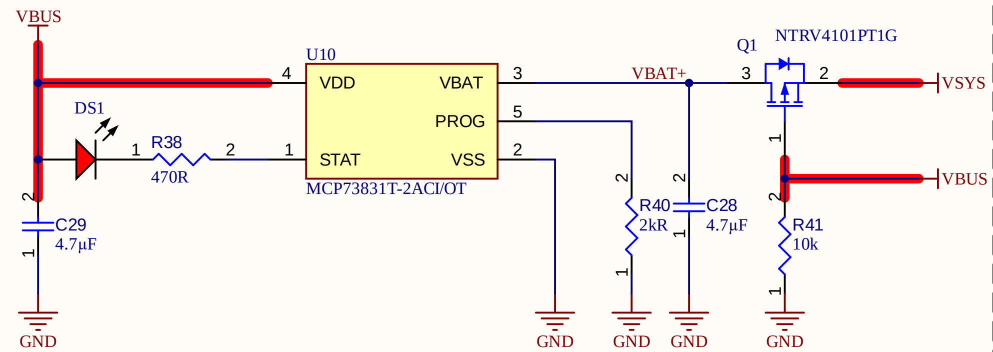


U8 sets ADC reference voltage to 2.5 V.

ADC reference voltage



Battery protection circuit



Battery protection circuit

U9 cutoff protection values:
OVP: 4.275V
UVP: 2.8V

J3 is designed to connect a single cell li-on battery.

R40 sets fast charge current to 500mA.

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TITLE: [7] PowerManagement.SchDoc	SIZE: A3; SCALE: 1:1
DOCUMENT NAME: [7] PowerManagement.SchDoc	SHEET: 7 OF 9
DOCUMENT REVISION: Not in version control	
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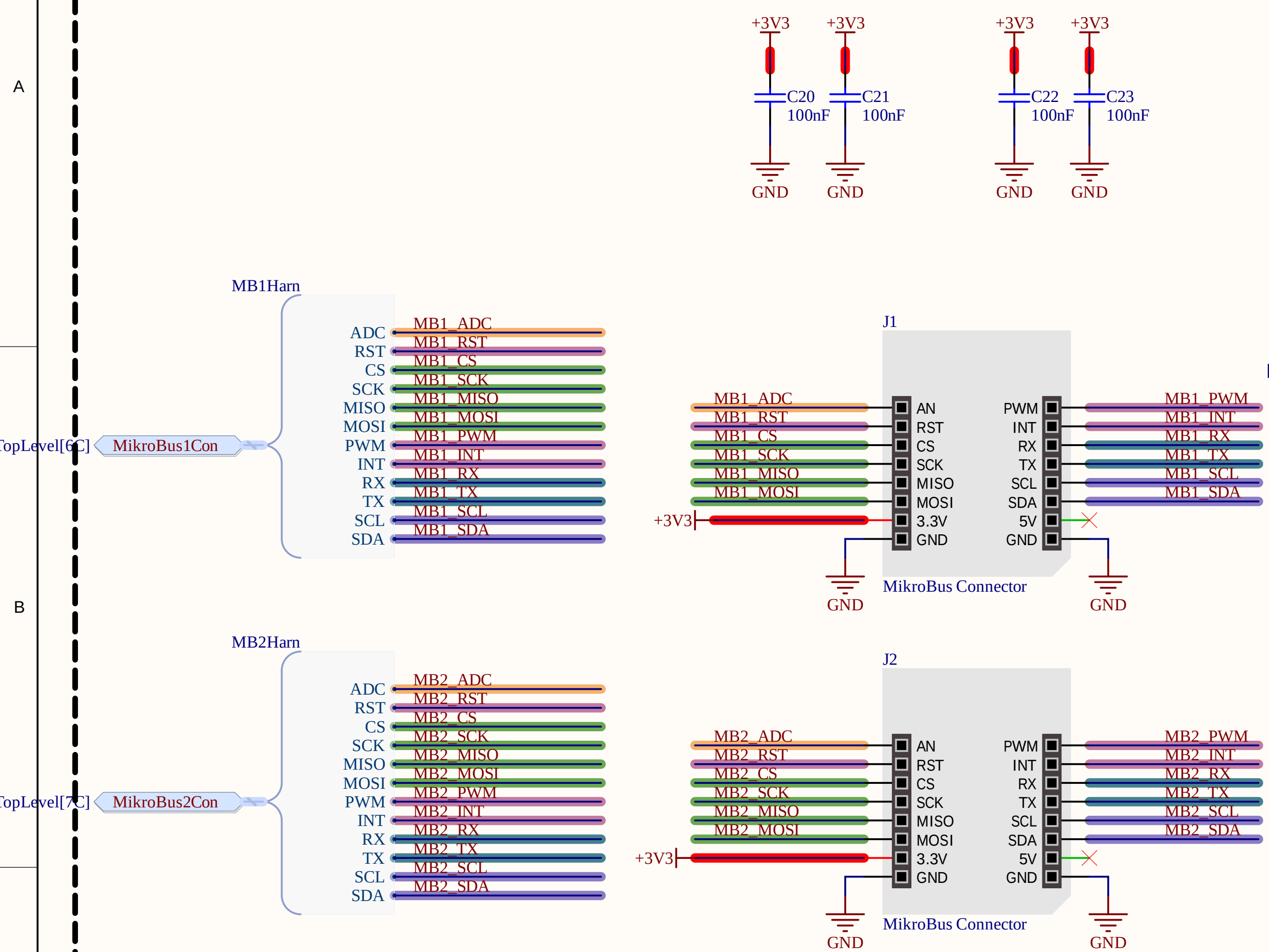


MikroBus connectors

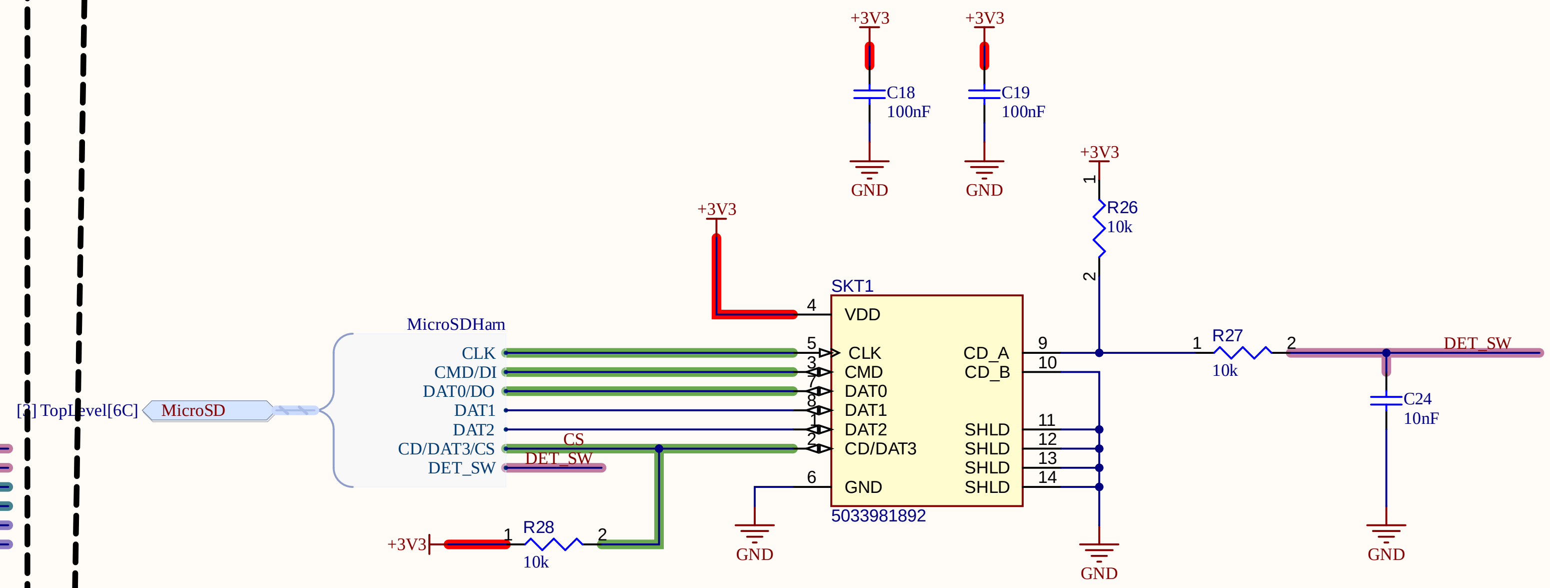
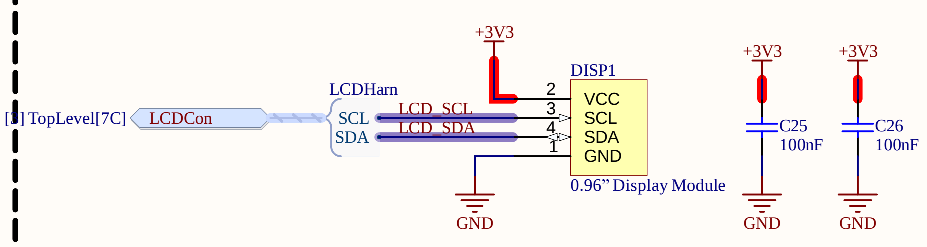


Table 2: microSD Card Pin Assignment

Pins	SD Mode			SPI Mode		
	Name	IO type [1]	Description	Name	IO Type	Description
1	DAT2	I/O /PP	Data Line[Bit2]	RSV		
2	CD/ DAT3	I/O/PP	Card Detect / Data Line[Bit3]	CS	I	Chip Select (neg true)
3	CMD	PP	Command/Response	DI	I	Data In
4	V_{dd}	S	Supply Voltage	V_{dd}	S	Supply Voltage
5	CLK	I	Clock	SCLK	I	Clock
6	V_{SS}	S	Supply voltage ground	V_{SS}	S	Supply voltage ground
7	DAT0	I/O /PP	Data Line[Bit0]	DO	O/PP	Data Out
8	DAT1	I/O /PP	Data Line[Bit1]	RSV	-	Reserved (*)

1) S: Power Supply, I: Input, O: Output, I/O: Bi-Directional, PP: IO Using Push-Pull Drivers
(*) These signals should be pulled up by the host side with 10-100K ohm resistance in SPI Mode. Do not use NC pins.

Micro SDCard Connector



External LCD Connector

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TITLE	SIZE	SCALE
[8] Conectors.SchDoc	A3	1:1

		SHEET:
DOCUMENT NAME:	[8] Conectors.SchDoc	8 OF 9
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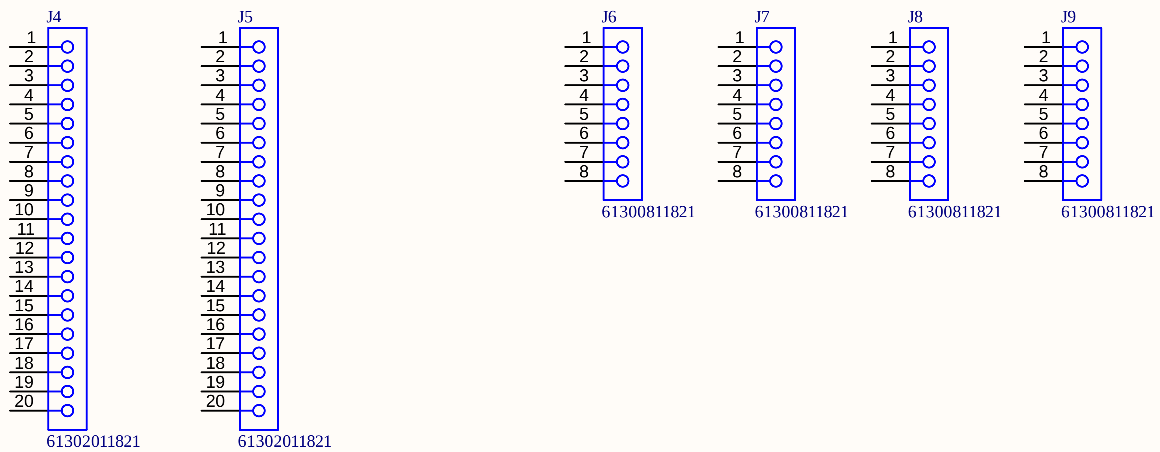

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TITLE [9] Mechanical.SchDoc		SIZE A3	SCALE 1:1
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